

Pre-Registered EFC Prediction: E_G Statistic from $\text{SO} \times \text{Euclid}$ Cross-Correlation

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Pre-registration date: 15 April 2026 **Data expected:** $\text{SO} \times \text{Euclid}$ DR2 ($\sim 2027\text{--}2028$)

Abstract. Energy-Flow Cosmology (EFC) predicts a specific enhancement of the gravitational slip statistic E_G relative to ΛCDM , arising from the simultaneous action of $\mu < 1$ (growth suppression) and $\Sigma > 1$ (lensing enhancement) derived self-consistently from a single variational action. We pre-register the quantitative prediction $E_G^{\text{EFC}}/E_G^{\text{GR}} = 1.086 \pm 0.012$, together with frozen pass/fail criteria, the dataset specification (Simons Observatory LAT $\times$ Euclid DR2), and explicit falsification conditions. All numerical values are derived from independently archived, timestamped sources. This document constitutes a sealed prediction prior to the availability of the targeted observational data.

1 Scope and Governance

This document records a frozen, pre-registered prediction for the E_G gravitational slip statistic, to be tested against the cross-correlation of Simons Observatory (SO) CMB lensing with Euclid DR2 galaxy clustering and weak lensing data.

Governance rules (frozen at pre-registration):

- No post-hoc adjustment of the predicted value or pass/fail window is permitted after this document is deposited.
- The EFC parameters ($\mu_0 = 0.94$, $\Sigma_0 = 1.05$, $\eta = 1.10$) are frozen in the Global Parameter Registry (DOI 31876324).
- Falsification is binding: if the kill criterion is met, the EFC gravitational slip sector is formally abandoned.
- All results — including failures — will be reported in the EFC Validation Ledger within one revision cycle.

What this prediction does *not* claim: This is a consistency test, not a discovery claim. A result consistent with the EFC prediction does not constitute confirmation of EFC; it constitutes non-rejection of the gravitational-slip sector.

2 Physical Basis

2.1 The E_G Observable

The E_G statistic combines CMB lensing, galaxy–galaxy lensing, and redshift-space distortions (RSD) to probe the ratio of the lensing potential to the velocity growth rate:

$$E_G(\ell) = \frac{C_\ell^{\kappa g}}{\beta C_\ell^{gg}}, \quad \beta = \frac{f}{b}, \quad (1)$$

where $C_\ell^{\kappa g}$ is the CMB-lensing×galaxy cross-power spectrum, C_ℓ^{gg} the galaxy auto-power, f the linear growth rate, and b the galaxy bias. In general relativity, $E_G^{\text{GR}} = \Omega_m(z)/f(z)$. In practice, the observable corresponds to a survey-weighted estimator and may differ from this idealised expression; all predictions refer to the pipeline-level definition used in SO × Euclid analyses. The reference quantity E_G^{GR} is evaluated within a standard Λ CDM background as used in survey pipelines. The EFC prediction is therefore defined relative to this operational baseline, not as a claim about the fundamental validity of the Λ CDM background itself. Both E_G^{obs} and E_G^{GR} are defined within the same survey pipeline and estimator framework to avoid inconsistencies arising from mixed definitions. For a modified gravity theory with $\mu \neq 1$ and $\Sigma \neq 1$:

$$E_G^{\text{MG}} = E_G^{\text{GR}} \times \frac{\Sigma}{\mu f/f_{\text{GR}}}, \quad (2)$$

making E_G the *orthogonal probe* to weak lensing alone: it is sensitive to the ratio Σ/μ , not to each separately. This relation holds in the quasi-static, sub-horizon regime relevant for the

adopted ℓ -range, and follows from the modified Poisson and lensing equations in that limit; see Bellini & Sawicki (2014) for the general EFT mapping.

2.2 EFC Prediction Derivation

The EFC Relativistic Action (DOI 31876324) derives from a single variational principle:

$$S_{\text{EFC}} = \int d^4x \sqrt{-g} \left[\frac{1}{2} M_{\text{Pl}}^2 F(\phi) R - \frac{1}{2} K(\rho) g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) - \lambda(\Box \phi - \Gamma(\rho)) \right]. \quad (3)$$

This action produces simultaneously:

$$\mu \approx 0.94 \quad (\text{growth suppression via entropy stiffness}), \quad (4)$$

$$\Sigma \approx 1.05 \quad (\text{lensing enhancement via flow anisotropy}), \quad (5)$$

$$\eta \approx 1.10 \quad (\text{gravitational slip, non-zero}). \quad (6)$$

From the Case B eigenmode analysis (DOI 31990194), the E_G ratio at $k_c \approx 0.05 h \text{ Mpc}^{-1}$ and $z \approx 0.5$ evaluates to:

$$\boxed{\frac{E_G^{\text{EFC}}}{E_G^{\text{GR}}} = 1.086 \pm 0.012} \quad (7)$$

where the uncertainty covers the survival valley $\mu \in [0.93, 0.96]$, $\Sigma \in [1.03, 1.07]$. The Bellini–Sawicki mapping (DOI 32011407) confirms $\alpha_T = 0$ exactly and derives $\alpha_M \propto S(a)$, $\alpha_B \propto dS/d \ln a$, establishing EFC compatibility with the Euclid EFT pipeline.

2.3 Why E_G is the Key Discriminant

Table 1: Discriminating power of E_G across competing models.

Model	$\mu < 1$	$\Sigma > 1$	$\eta \neq 1$	$E_G^{\text{obs}}/E_G^{\text{GR}} > 1$
EFC	✓	✓	✓	✓
Massive neutrinos	✓			
$f(R)$ gravity				
Horndeski (general)	possible	possible		possible
$w_0 w_a \text{CDM}$				

No currently established model listed in Table 1 produces $\mu < 1$ and $\Sigma > 1$ and $\eta \neq 1$ simultaneously from a single action. A detection of $E_G^{\text{obs}}/E_G^{\text{GR}} \approx 1.09$ would provide strong evidence in favour of EFC-class theories over all alternatives in the table.

3 Dataset Specification

Table 2: Pre-registered dataset specification.

Element	Specification
CMB lensing	Simons Observatory LAT κ reconstruction, $\ell \in [100, 2000]$
Galaxy sample	Euclid DR2 photometric catalogue, $z \in [0.3, 0.8]$, 5 tomographic bins
ℓ -range	$\ell \in [50, 500]$ (linear regime; $k < 0.2 h \text{ Mpc}^{-1}$)
Pivot scale	$k_c = 0.05 h \text{ Mpc}^{-1}$, $z_{\text{eff}} = 0.5$
Bias treatment	Survey-provided NLA + linear bias marginalisation
Covariance	Survey-provided covariance matrices; no EFC nuisance parameters
Expected date	Euclid DR2 + SO cross-correlation $\sim 2027\text{--}2028$

Note on tomographic binning: Euclid tomographic window functions are expected to reduce the intrinsic signal amplitude by $\sim 30\text{--}40\%$. The pass/fail window in Section 4 is defined for the *tomography-convolved* observable; no correction is applied post-hoc.

4 Pass/Fail Criteria

Table 3: Pre-registered pass/fail criteria (frozen, no post-hoc revision).

Criterion	Threshold	Consequence	Regime
KC-EG-PASS	$E_G^{\text{obs}}/E_G^{\text{GR}} \in [1.03, 1.14]$ at $> 1\sigma$	Consistent with EFC prediction	FA4 supported
KC-EG-FAIL	$E_G^{\text{obs}}/E_G^{\text{GR}} = 1.00 \pm 0.01$ at $> 3\sigma$	EFC gravitational-slip sector falsified	FA4 triggered
KC-EG-INC.	Result between pass and fail windows	No action; await Euclid DR3	—

Kill criterion FA4 (from the Stage-IV Roadmap, DOI 31970904): if $|\eta - 1| < 0.01$ is established at $> 3\sigma$ from the $\text{SO} \times \text{Euclid}$ EG measurement, the EFC perturbation sector is formally falsified and the result is permanently documented in the Validation Ledger.

Note on pass window width: The pass window $[1.03, 1.14]$ is intentionally broader than the 1σ prediction interval to account for survey window functions, tomographic convolution, and estimator-level effects that are not modelled post-hoc. This is a pre-registered design choice, not a soft target.

5 Regime Declaration and Systematics

RCMP regime declaration. Following the Regime-Consistent Measurement Principle (DOI 31222900), the three components of E_G are classified by regime: CMB lensing (L1 linear, large-scale); galaxy clustering (L1/L2 boundary, bias-dependent); RSD (L1 growth sector). This

declaration makes explicit a consistency condition that is typically implicit in standard analyses. Cross-regime mixing is declared here, not corrected for post-hoc. All predictions are evaluated at the intersection of these regimes at $k_c = 0.05 h \text{ Mpc}^{-1}$, $z_{\text{eff}} = 0.5$, where all three components remain in the linear regime. The prediction is stated at a representative pivot scale rather than as a full functional form, to maintain strict pre-registration without introducing additional model freedom.

Table 4: Known systematics and EFC response.

Systematic	Effect on E_G	EFC robustness
Intrinsic alignments	Shifts $C_\ell^{\kappa g}$ amplitude	Marginalised; NLA model per survey spec
Photo- z errors	Dilutes tomographic signal	Scale cuts at $k < 0.2 h \text{ Mpc}^{-1}$ applied
Galaxy bias	Enters via $\beta = f/b$	Cancels in E_G ratio to leading order
Magnification bias	Sub-percent at $\ell < 500$	Below signal threshold
Baryonic feedback	Confined to $k > 0.5 h \text{ Mpc}^{-1}$	Outside pre-registered ℓ -range
Massive neutrinos	Suppress μ but $\eta = 1$	Discriminated by $\eta \neq 1$ measurement
RSD modelling	Affects $\beta = f/b$ estimation	Mitigated via survey pipeline; cancels partially in E_G ratio
CMB lensing reconstruction	Bias in $C_\ell^{\kappa g}$ from map filtering	Mitigated via standard SO pipeline null tests and simulations

6 Reproducibility

- **RCMP** (regime-consistent measurement principle): DOI 31222900
- **EFC Relativistic Action** (derivation of μ, Σ, η): DOI 31876324
- **Case B eigenmode** ($E_G^{\text{EFC}}/E_G^{\text{GR}} = 1.086$): DOI 31990194
- **Bellini–Sawicki mapping** (hi_class compatibility): DOI 32011407
- **Stage-IV Roadmap** (KC4 kill criterion): DOI 31970904
- **EFC Pre-Registered Predictions** (sealed parameter set): DOI 32010399
- **GitHub repository**: github.com/supertedai/EFC
- **Validation Ledger**: EFC Validation Ledger

All code and data are version-controlled. The git commit hash at the time of analysis will be recorded in the Validation Ledger entry. No code modification is permitted after unblinding.

Final statement. This document constitutes a complete, frozen pre-registration for the EFC E_G prediction prior to the availability of SO $\times$ Euclid DR2 data. The predicted value $E_G^{\text{EFC}}/E_G^{\text{GR}} = 1.086 \pm 0.012$, the pass/fail window, the dataset specification, and the falsification conditions are not subject to modification. Pre-registration date: 15 April 2026.

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